Generate the FPGA Programming Bitstream

This guide explains how to Generate the full `k5_xbox` FPGA programming bitstream in the RC cloud environment with your `sudx_basic` accelerator integrated, and download it to your laptop for FPGA programming.

Prerequisites - Synthesizing the Accelerator Top-Level `sudx_basic`

Before compiling the complete FPGA system, your accelerator must first pass the standalone feasibility check, as described in the following guide: **ddp26_qsyn_xlr_guide.pdf**

Although this step is not required to compile the full FPGA system including the accelerators, as described below, it runs much faster than a full FPGA compilation and is therefore very useful for performance feedback and debug iterations.

check synthesis of your top-level `sudx_basic` module.

```
cd $MY_K5_XLRS/sudx_basic
qsyn_xlr  sudx_basic -all
```

Full System FPGA Compilation

In your RC cloud environment, open a new terminal and navigate to the FPGA bitstream generation folder:

```
cd $MY_K5_PROJ/hw/gen_fpga
```

Run the full FPGA compilation.

Note: This process may take more than ten minutes, depending on the complexity of your accelerators.

```
comp_fpga sudx_basic
```

The `comp_fpga` utility is similar to `qsyn_xlr`, but it is designed to compile the complete FPGA system, apply the necessary integration adjustments, and generate the final FPGA programming bitstream.

You may optionally specify flags such as `-syn`, `-fit`, or `-all` (the default is `-all`).

- Use `-syn` for faster synthesis-only runs.
- Most of the compilation time is commonly spent during the fitting phase.

Generated Bitstream Files and Downloads

Upon successful completion, the following FPGA programming bitstream file will be generated:

```
# For now just verify this file exists by:
ls $MY_K5_PROJ/hw/gen_fpga/prog_files/k5_xbox_sudx_basic.sof
```

Download this `.sof` file to your laptop. It will be used to program the FPGA board, and locate it on your laptop environment at `$MY_K5_PROJ/fpga_prog_files/k5_xbox_sudx_basic.sof`

In addition, download the accelerator application C source code folders both `sudx_basic` and `sud_shared` for compilation (and possible modification) within your FPGA host Windows environment at: `$MY_K5_PROJ/sw/apps`

To simplify access, your cloud desktop contains a folder shortcut named `k5_xbox_links`

Within this folder, you will find convenient links to:

- Software application directories
- FPGA programming files

When accessing *My Files* from your laptop, you may use these links instead of manually navigating to the full file paths.

Running the Accelerated Application on the FPGA

To make sure your Windows environment is up to date with the latest environment updates, perform the following steps **once** in a Git Bash terminal:

```
cd $K5X_WIN/k5_xbox_fpga_win
git pull
```

After copying `sudx_basic.sof` from the cloud to the laptop, and copying the application SW folders as described above, you can attempt to run the accelerated application on the FPGA.

First, set up the FPGA board as described in the previous guide, and then execute the following commands:

```
set_k5_terminal
prog_fpga sudx_basic
launch_k5_app sudx_basic -asl sud_shared

# To run with Acceleration ON
launch_k5_app sudx_basic -asl sud_shared -ccd1 XON
```

The application should produce exactly the same functional results as in simulation, but run significantly faster on the FPGA.